



Review

Embryonal Fyn-associated substrate (EFS) and CASS4: The lesser-known CAS protein family members



Alexander Deneka^{a,b}, Vladislav Korobeynikov^{a,c}, Erica A. Golemis^{a,*}

^a Molecular Therapeutics Program, Fox Chase Cancer Center, Philadelphia, PA, 19111, United States

^b Kazan Federal University, 420000, Kazan, Russian Federation

^c Novosibirsk State University, Medical Department, 630090, Novosibirsk, Russian Federation

ARTICLE INFO

Article history:

Received 14 May 2015

Accepted 23 June 2015

Available online 26 June 2015

Keywords:

EFS
CASS3
SIN
CASS4
HEPL
CAS family protein
Inflammation
Signal transduction
Prognosis

ABSTRACT

The CAS (Crk-associated substrate) adaptor protein family consists of four members: CASS1/BCAR1/p130Cas, CASS2/NEDD9/HEF1/Cas-L, CASS3/EFS/Sin and CASS4/HEPL. While CAS proteins lack enzymatic activity, they contain specific recognition and binding sites for assembly of larger signaling complexes that are essential for cell proliferation, survival, migration, and other processes. All family members are intermediates in integrin-dependent signaling pathways mediated at focal adhesions, and associate with FAK and SRC family kinases to activate downstream effectors regulating the actin cytoskeleton. Most studies of CAS proteins to date have been focused on the first two members, BCAR1 and NEDD9, with altered expression of these proteins now appreciated as influencing disease development and prognosis for cancer and other serious pathological conditions. For these family members, additional mechanisms of action have been defined in receptor tyrosine kinase (RTK) signaling, estrogen receptor signaling or cell cycle progression, involving discrete partner proteins such as SHC, NSP proteins, or AURKA. By contrast, EFS and CASS4 have been less studied, although structure–function analyses indicate they conserve many elements with the better-known family members. Intriguingly, a number of recent studies have implicated these proteins in immune system function, and the pathogenesis of developmental disorders, autoimmune disorders including Crohn's disease, Alzheimer's disease, cancer and other diseases. In this review, we summarize the current understanding of EFS and CASS4 protein function in the context of the larger CAS family group.

© 2015 Elsevier B.V. All rights reserved.

Abbreviations: 5-HT, 5-hydroxytryptamine, serotonin; A549, non-small cell lung cancer cell line; AD, Alzheimer's disease; ADPKD, autosomal dominant polycystic kidney disease; Aire, autoimmune regulator; AP-1, activator protein 1; APP, amyloid precursor protein; AREB6, ZEB1; ATF, activating transcription factor; AURKA, aurora A kinase; BCAR1, breast cancer anti-estrogen resistance 1; BE1, non-small cell lung cancer cell line; BRAF, v-Raf murine sarcoma viral oncogene homologue B; c-CRK, cellular CRK; c-RAF, cellular RAF; c-SRC, cellular SRC kinase; C/EBP α , CCAAT/enhancer-binding protein alpha; C20orf32, chromosome 20 open read frame 32; C3G, RAPGEF1, CRK SH3-binding GNRP; CAKb, PTK2B, cell adhesion kinase; CAS, Crk-associated substrate; Cas-L, NEDD9, Crk-associated substrate, lymphocyte type; CASS3, Crk-associated substrate 3; CASS4, Crk-associated substrate 4; CBLN4, cerebellin-4; CCAAT, DNA sequence; CD4/8, cluster of differentiation 4/8; CDCP1, CUB domain-containing protein 1; CGS, Canadian Consortium for Genetic Studies; CHOP-10, C/EBP homologous protein 10; CHS, Chediak-Higashi syndrome; CHS1, Chediak-Higashi syndrome protein 1; CIZ, Cas-interacting zinc finger protein; CRK, v-Crk avian sarcoma virus CT10 oncogene homologue; Crk-L, Crk-like protein; CSTF1, cleavage stimulation factor subunit 1; DAG, diacylglycerol; DCAS, Drosophila Crk-associated substrate; DECODE, DECipherment Of DNA Elements; ECRG4, esophageal cancer-related gene 4 protein, Augurin; EFS, embryonal Fyn-associated substrate; EFS1, embryonal Fyn-associated substrate 1; EFS2, embryonal Fyn-associated substrate 2; ERK, extracellular-signal-regulated kinase; FAK, focal adhesion kinase; FLNC, filamin-C; FRNK, FAK-related non-kinase; GAS, group A streptococci; GATA-3, transcription factor binding GATA DNA sequence; GEF, guanine nucleotide exchange factor; GEP, gene expression profile; GMS, Gene Modifier Study; GRChB38p2, Genome Reference Consortium Human Build 38 patch release 2; GWAS, genome-wide association study; H1299, non-small cell lung cancer cell line; HEF1, NEDD9, human enhancer of filamentation 1; HEFL, HEF like protein; HEFS, human EFS; HEPL, HEF1-EFS-p130Cas-like protein; HOP-62, lung adenocarcinoma cell line; IGAP, International Genomics of Alzheimer's Disease Project; IL-2, interleukin 2; IP3, inositol three phosphate; KCNMA1, potassium large conductance calcium-activated channel, subfamily M, alpha member 1; KGF, keratinocyte growth factor; KODA-PC, 9-keto-12-oxo-10-dodecanoic acid ester of 2-lyso-phosphocholine; LCR-F1, Locus Control Region-Factor 1; LNCaP, prostate cancer cell line; LTE, non-small cell lung cancer cell line; LYST, lysosomal trafficking regulator; MAX1, Myc-associated factor X; MC3R, melanocortin receptor 3; MHC I/II, major histocompatibility complexes I/II; MSI, microsatellite instability; MSS, microsatellite stable; mTEC, medullar thymus epithelial cells; Nck, non-catalytic region of tyrosine kinase adaptor protein; NEDD9, neural precursor cell expressed, developmentally down-regulated 9; NF- κ B, nuclear factor kappa-light-chain-enhancer of activated B cells; NF- κ B1, nuclear factor NF-kappa-B p105 subunit; NFAT, nuclear factor of activated T-cells; NFE2-L1, nuclear factor, erythroid 2-like 1; NSCLC, non-small cell lung cancer; NSP, novel SH2 containing protein; Oct-6, octamer transcription factor 6; OR, odds ratio; PBMC, peripheral blood mononuclear cell; PC, prostate cancer; PDLIM4, PDZ and LIM domain 4; PIP2, phosphoinositide diphosphate; PITX2, pituitary homeobox 2; PLC- γ , phosphoinositide phospholipase C gamma; POU, Pit-Oct-Unc; POU3F1, POU domain, class 3, transcription factor 1, aka Oct-6; PTK2B, protein tyrosine kinase 2B; PTP-PEST, protein tyrosine phosphatase containing C-terminal PEST motif; PTP1B, protein tyrosine phosphatase 1B; PTPase, protein tyrosine phosphatase; Pyk2, PTK2B, protein tyrosine (Y) kinase; RAFTK, PTK2B, related adhesion focal tyrosine kinase; RAP1, Ras-proximate-1 or Ras-related protein 1; RAPGEF1, Rap guanine nucleotide exchange factor 1; RHD, rheumatoid heart disease; RPTP α , receptor–protein–tyrosine phosphatase alpha; RTK, receptor tyrosine kinase; SH2, Src homology 2 domain; SH3, Src homology 3 domain; SHC, Src homology 2 domain containing; Sin, Src interacting or signal integrating protein; SNP, Single nucleotide polymorphism; TCR, T-cell receptor; TEC, thymus epithelial cell; TNM, tumor, nodules, metastases scale; Trask, CDCP1, transmembrane and associated with src kinases; TSS, Twins & Sibs Study; UCSC, University of California Santa Cruz; UM, uveal melanoma; v-RAF, virus-induced rapidly accelerated fibrosarcoma; ZEB1, zinc finger E-box binding homeobox 1.

* Corresponding author at: Fox Chase Cancer Center, 333 Cottman Ave., Philadelphia, PA 19111, United States.

E-mail address: Erica.Golemis@fccc.edu (E.A. Golemis).

Contents

1.	Introduction	26
2.	Identification of EFS and CASS4	26
3.	Gene and mRNA expression for EFS and CASS4	27
4.	Protein structure	27
5.	EFS, CASS4, and canonical CAS protein functions	28
6.	EFS and immune system function	29
7.	Implication of EFS and CASS4 in other disorders	31
7.1.	Alzheimer's disease	31
7.2.	Cystic fibrosis	32
7.3.	Thrombosis	32
7.4.	Cancer	32
8.	Conclusions	34
	Acknowledgments	34
	References	34

1. Introduction

The CAS (CRK-associated substrate) family of adaptor proteins consists of four members: BCAR1 (also known as p130Cas and CASS1) (Sakai et al., 1994a), NEDD9 (also known as HEF1, Cas-L, and CASS2) (Law et al., 1996; Minegishi et al., 1996), EFS/SIN/CASS3 (Ishino et al., 1995) and HEPL/CASS4 (Singh et al., 2008). Each of these proteins contains specific binding and recognition sites allowing them to serve as scaffolds to assemble larger signaling complexes, and lacks known enzymatic activity. Intensive study predominantly of BCAR1 and NEDD9 in the 1990s led to the early development of a paradigm for the function of this protein group (reviewed in O'Neill et al., 2000). In brief, engagement of integrins at the cell surface causes activation of focal adhesion kinase (FAK) or its paralog PTK2B (also known as RAFTK or Pyk2), which binds and phosphorylates a CAS protein. Subsequently, a SRC family kinase binds a site created by the FAK-phosphorylation event and hyperphosphorylates CAS, creating binding sites for CRK and associated proteins that reorganized the actin cytoskeleton in a manner that influences integrin-mediated attachment, cell spreading, and migration. Disruption of the function of these complexes led to anoikis (detachment-induced cell death), while overexpression or hyperphosphorylation of CAS proteins accompanies and supports cell transformation.

In this context, a literature addressing the paralogous EFS and CASS4 family members has been slower to emerge. In general, an advantage of studies in lower eukaryotes is the opportunity to dissect protein function in a signaling environment in which few paralogues are present. In mammals, it is often challenging to distinguish redundant versus unique signaling features, and therefore establish the biological importance of each gene within expanded protein families. Given the near-ubiquitous expression of BCAR1 in particular (Barrett et al., 2013), and the abundant expression of NEDD9 in many tissues (Minegishi et al., 1996; Fashena et al., 2002), identifying discrete roles for other CAS family members is not trivial. However, there is a growing number of cases where genes that were thought to have predominantly redundant function were subsequently shown to take part in distinct feedback loops and synergistic signaling, or to have partially opposing function that can have important biological consequences and influence clinical outcomes in the case of disease. As one example, clinical effectiveness of RAF inhibitors in the context of V600E mutant (constitutively active) versus wild type BRAF is influenced strongly by the functionality of the paralogous CRAF protein, based on heterodimerization and transactivation processes specific to each of the two proteins (Poulikakos et al., 2011). Such examples reinforce the importance of understanding the role of individual paralogs.

In the CAS family, as discussed below, all family members share common structural features and have at least some similar functions in cell signaling, related to the paradigm noted above. However,

individual roles have emerged for the better-studied BCAR1 and NEDD9 proteins that differentiate part of their function from the remainder of the CAS group. For example, BCAR1 expression regulates tamoxifen resistance in breast cancer (van Agthoven et al., 2009), and plays a crucial role in cardiovascular system and liver development (Tazaki et al., 2010) and osteoclast-dependent bone resorption (Nagai et al., 2013). BCAR1 supports myelin formation by oligodendrocytes in the central nervous system (Gonsior et al., 2014), acts as a downstream effector of serotonin (5-HT) signaling in the process of vasoconstriction (Ogden et al., 2006), and serves as an important mediator for several infectious diseases including infection by *Salmonella typhimurium* (Shi and Casanova, 2006), *Plasmodium falciparum*-associated malaria (Davis et al., 2012), pathogenic *Yersinia* (Bruce-Staskal et al., 2002; Weidow et al., 2000), and Kaposi's sarcoma-associated herpesvirus (Bandyopadhyay et al., 2014). NEDD9 is known to regulate neural crest cells migration during embryogenesis, which is crucial for proper neural system development (Aquino et al., 2009), functions at centrosomes to activate the Aurora-A (AURKA) mitotic kinase (Pugacheva and Golemis, 2005), and at the basal body to activate AURKA in causing disassembly of the primary cilia (Pugacheva et al., 2007). Based on these non-canonical roles, NEDD9 function has recently been shown to modulate pathogenesis of the ciliopathy autosomal dominant polycystic kidney disease (ADPKD) (Nikonova et al., 2014a).

In contrast, the literature on EFS and CASS4 has been slower to develop. Nevertheless, over the past several years, a growing number of studies have begun to define similarities and differences between these and other CAS proteins, and to implicate EFS and CASS4 as causative factors or effectors for clinically important human diseases, including developmental disorders, neurodegenerative syndromes, autoimmune disorders, and cancer. In this review, we summarize the current status of the literature on these two proteins, their structure, functions and roles in signaling transduction of signaling pathways.

2. Identification of EFS and CASS4

EFS (Embryonal Fyn-associated Substrate), also known as SIN (Src Interacting or Signal Integrating protein) was discovered in two independent studies, Ishino et al. (1995) and Alexandropoulos and Baltimore (1996). Ishino and colleagues performed a cDNA library screening of a mouse embryonal library for proteins containing SH3-interacting domains to identify EFS, while Alexandropoulos screened a mouse embryonal library and looked for proteins interacting with the SRC SH3 domain, leading to the designation SIN.

CASS4, the last described member of the CAS family, was detected by Singh et al. (2008) over a decade after the description of the other family members. CASS4 was identified following in silico screening

of databases describing expressed sequence tags from an evolutionarily diverse group of organisms, using the mRNAs for the three previously defined CAS proteins as templates. Subsequently, Singh et al. cloned and characterized the CASS4 gene, originally assigning the name HEPL (HEF1-EFS-p130Cas-like) for similarity to the other three defined CAS genes.

3. Gene and mRNA expression for EFS and CASS4

The EFS gene is localized to chromosome 14q11.2, with genomic coordinates 14: 23356400–23365633 on the reverse strand in GRChB38p2 (EntrezGene, 2015a). The chromosomal location of the CASS4 gene is 20q13.31, with genomic coordinates of 20: 56411548–56459340 on the forward strand in GRChB38p2 (EntrezGene, 2015b).

Relatively little work has been done to directly study the transcriptional regulation of EFS and CASS4. Initial studies profiling EFS mRNA indicated broad expression, with maximal levels in the placenta, the embryonal central nervous system, heart, testes and lungs (Ishino et al., 1997). EFS expression in the thymus and lymphocytes is functionally important for T cell maturation and prevention of autoimmunity, discussed below (Donlin et al., 2002, 2005; Danzl et al., 2010). A screen for implantation-related genes regulated by 17 β -estradiol and progesterone found this regimen downregulated EFS mRNA in explants of the late proliferative phase endometrium, suggesting hormone responsiveness (Dassen et al., 2007). The initial publication describing CASS4 used RT-PCR to demonstrate that CASS4 mRNA is most highly expressed in spleen and lung among normal tissues, and is highly expressed in ovarian and leukemia cell lines (Singh et al., 2008).

Some further hints as to the transcriptional control of these genes can be gleaned from SABiosciences' DECODE database, which is based on UCSC Bioinformatics Genome Browser (UCSC). This resource proposes several transcriptional regulators for EFS based on consensus binding sites in its promoter region for ATF (Activating transcription factor), NF- κ B, NF- κ B1, GATA-3, C/EBP α (CCAAT/enhancer-binding protein alpha), glucocorticoid receptors α and β , and p53. Transcription factor binding motifs in the CASS4 promoter are similar to those for EFS. They include NF- κ B, p53, LCR-F1 (NFE2-L1, nuclear factor, erythroid 2-like1), MAX1, C/EBP α , CHOP-10 (C/EBP homologous protein 10), POU3F1 (POU domain, class 3, transcription factor 1, aka Oct-6), AREB6 (ZEB1, zinc finger E-box binding homeobox 1). These are compatible with regulated expression in lymphocytes in relation to differentiation and inflammation (Alexandropoulos et al., 2003; Tikhmyanova et al., 2010a), hormonal regulation (Nikonova et al., 2014b), and deregulation in cancer (Tornillo et al., 2014; Wallez et al., 2012).

There are three transcript variants for EFS in humans. Isoform 1 contains 6 exons and encodes the full-length protein (561 amino acids); isoform 2 contains 5 exons and encodes a shorter protein (468 amino acids); and isoform 3 contains 6 exons and encodes the shortest protein (392 amino acids). In humans, four transcript variants of CASS4 are known. The first and second variants contain 7 exons each and encode the same full-length protein isoform a (786 amino acids, considered the major isoform). The third variant contains 6 exons and encodes a

shorter isoform b (732 amino acids), and the fourth variant contains 5 exons and encodes the shortest isoform c (349 amino acids).

4. Protein structure

The evolutionary divergence of the CAS proteins family members is discussed by Singh et al. in detail (Singh et al., 2008). There are no CAS homologues in single cell eukaryotes and nematodes such as *Caenorhabditis elegans*. A single ancestral member, DCAS, is found in *Drosophila melanogaster* that has common structural characteristics and functions with other proteins within the group (Tikhmyanova et al., 2010b). The characteristic domains and motifs are summarized in Table 1 and graphically represented in Fig. 1.

All CAS proteins maintain a single N-terminal and highly conserved SH3-domain, which allows binding to proline-rich motif containing proteins (Ishino et al., 1995). This encompasses amino acids 5–68 and 11–73 for EFS and CASS4 respectively. This SH3 domain is highly conserved, with 81% of overall similarity between the CAS family proteins. For EFS and CASS4, the SH3 domains are the most highly conserved: for example, the murine and human EFS SH3 domains are 100% identical, while the remainder of the coding sequences are only 78% identical (Ishino et al., 1997), which indicates an important functional role. Similarly, for CASS4, the murine and human SH3 domains are 87.3% identical, while the remainder of coding sequence has 59.8% identity. Important binding partners for CAS family members for this region include FAK (Minegishi et al., 1996), PTK2B (Astier et al., 1997), C3G (Kirsch et al., 1998), PTP-PEST (Garton et al., 1997), PTP1B (Liu et al., 1998), CIZ (Nakamoto et al., 2000) and FRNK (Harte et al., 1996). Singh et al. used a yeast two-hybrid approach to directly demonstrate an interaction between the CASS4 SH3 domain and FAK. EFS has also been shown to interact with FAK, with an affinity comparable to p130Cas/BCAR1 and NEDD9 (Ohba et al., 1998). PTP-PEST, a soluble protein tyrosine phosphatase that is ubiquitously expressed in mice both during embryonic development and in adult tissues, opposes FAK and PTK2B activity, as it dephosphorylates PTK2B, FAK and CAS family members, among other proteins (Davidson and Veillette, 2001). The PTP-PEST proline-rich sequence ³³²PPKPPR³³⁷ has been shown to interact directly with the SH3 domain of EFS (Cote et al., 1998). This set of binding partners for a highly conserved domain reflects the fundamental role(s) of this SH3 domain in mediating integrin-dependent cell adhesion.

Carboxy-terminal to the SH3 domain, CAS proteins contain a “substrate domain” (amino acids 69–350 for human EFS and 74–372 for human CASS4) with multiple repeats of specific sequences (YxxP) which are phosphorylated on tyrosines by SRC or SRC-related kinases, creating binding motifs for SH2 domains-containing signaling proteins (Songyang et al., 1993). Both EFS and CASS4 have a limited number of potential SH2 binding sites (estimated at 9 and 10, respectively), in contrast to BCAR1 and NEDD9, which have 20 and 18 of these motifs (Singh et al., 2008). Important interactive partners binding this region include Crk1/2 and Crk-L, SH2- and SH3 domain-containing adaptor proteins that are important for regulation of cellular motility and migration (Law et al., 1996; Singh et al., 2008; Ishino et al., 1997; Sakai et al.,

Table 1
Interactive partners defined for association with specific domains of EFS or CASS4.

Domain	Protein	Amino acids	Length	Interactors	Refs.
SH3	EFS	5–68	64	FAK, PTK2B, C3G, PTP-PEST	Ohba et al. (1998), Xing et al. (2000), Cote et al. (1998) Singh et al. (2008)
	CASS4	11–73	63	FAK	
SH2-binding	EFS	69–350	282	Crk1/2, CrkL	Alexandropoulos and Baltimore (1996), Wu et al. (2007) Brehme et al. (2009), Zahedi et al. (2008)
	CASS4	72–372	301	CrkL, PKA/PKG	
Serine rich	EFS	351–488	138		Alexandropoulos and Baltimore (1996), Donlin et al. (2002), Wu et al. (2007) Singh et al. (2008)
	CASS4	373–429	57		
C-terminal	EFS	489–561	73	SRC, Fyn	
	CASS4	430–786	357	SRC	

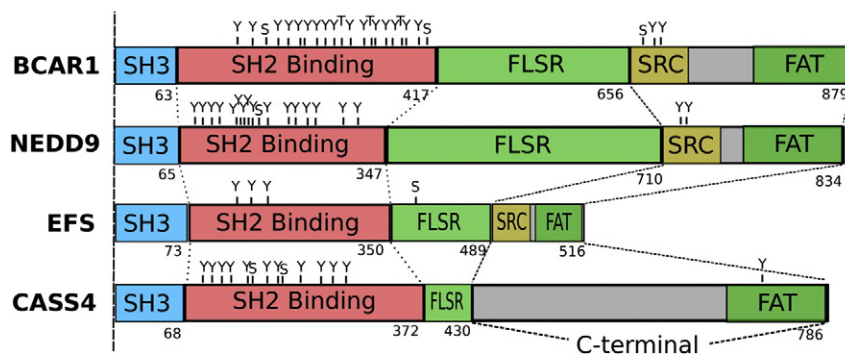


Fig. 1. Schematic domain structure and phosphorylation sites comparison of the CAS family proteins. Shown are the longest isoforms for the members of CAS protein family. SH3 = SRC homology 3 binding domain; SH2 binding = tyrosine phosphosite-enriched domain containing binding sites for partners with SH2 domains; FAT = focal adhesion targeting domain (the central serine rich region (FLSR) of the CAS proteins although has the structure of FAT domain); SRC = SRC family kinase-binding region, contains binding sites for both SRC SH2 and SH3 domains in BCAR1 and EFS, and only SRC SH2 domain in NEDD9. Domain amino acid lengths were obtained from uniprot.org database, indicated in numbering below each image. Phosphorylation sites are designated as follows: Y = tyrosine; S = serine; T = threonine; and were obtained from phosphosite.org.

1994b); and C3G, a guanine nucleotide exchange factor (GEF) for RAP1 (Xing et al., 2000). Interactions with CRK and C3G subsequently promote reorganization of the actin cytoskeleton (Tikhmyanova et al., 2010b). Direct binding has been identified between CRKL and CASS4 (Brehme et al., 2009; Hsia et al., 2003), while EFS has been shown to bind Crk1/2 (Alexandropoulos and Baltimore, 1996) and CrkL (Wu et al., 2007). Studies of BCAR1 have shown that YxxP sequences in this region become more available to phosphorylation by SRC upon mechanical stretch of the protein (Sawada et al., 2006) which implies a role of CAS proteins as direct intracellular mechanoreceptors mediating cellular response to mechanical forces, especially considering that all four CAS proteins have been reported to localize at focal adhesions (Law et al., 1996; Singh et al., 2008; Xing et al., 2000; Petch et al., 1995).

Carboxy-terminal to the substrate domain is a serine-rich domain encompassing a 4 α -helix bundle (amino acids 351–488 for human EFS and 373–429 for CASS4). Although primary amino acid sequence shows considerable divergence versus other CAS family members in this region, structural analysis predicts that this bundle has a highly conserved fold and provides a docking site for family members. Direct interacting partners for this region have not yet been identified for CASS4 and EFS.

The C-terminal domain (around 120 amino acids in length) is the second most strongly conserved among family members (after the SH3 domain) at both the level of the primary amino acid sequence and predicted fold (Singh et al., 2008). The overall similarity for this sequence is estimated at 51%. This domain has been shown to mediate to provide homo-dimerization for NEDD9 (Law et al., 1999), and is involved in interactions with some important binding partners including SRC for CASS4 (Singh et al., 2008) and EFS (Alexandropoulos and Baltimore, 1996) and Fyn kinase for EFS (Donlin et al., 2002). For BCAR1 and NEDD9, important interactors in this region also include the NSP proteins, which provide connections to receptor tyrosine kinase signaling (Wallez et al., 2012); interactions of EFS and CASS4 in this region have been minimally probed. In study of mechanosensation by BCAR1 discussed above (Sawada et al., 2006), anchoring of the protein based on interactions between FAK at the SH3 domain and a complex involving SRC-family kinases at the C-terminal domain provides the two poles between which the substrate domain is stretched to become active.

In terms of short motifs, all CAS proteins except CASS4 contain a flanking YDYVHL sequence. Typically, phosphorylation of this motif by FAK or PTK2B creates a binding site for the SH2 domain of a SRC-family protein (Tachibana et al., 1997), which then hyperphosphorylates the substrate domain. In contrast to other family members, EFS and BCAR1 have also been shown to contain two and one proline-rich motifs, respectively (Alexandropoulos and Baltimore,

1996), which may enhance the binding and phosphorylation by SRC family kinases, based on interactions with the SRC SH3 domain (Pellicena and Miller, 2001).

Despite the reduced overall sequence similarity of CASS4 to other family members, molecular modeling analysis performed by Singh et al. (2008) using p130CAS/BCAR1 structures as templates suggested an almost identical fold between CASS4 and p130CAS/BCAR1 within their SH3 domains, and substantial similarity within 432–591 residues of CASS4 and 449–610 residues of p130Cas/BCAR1 at the level of secondary and tertiary structures. Also, the similar periodicity of α -helices and β -sheets in both CASS4 and p130Cas/BCAR1 provides another confirmation for the idea of well-conserved structures within the family members.

5. EFS, CASS4, and canonical CAS protein functions

The canonical signaling activities of EFS and CASS4 proteins are outlined in Fig. 2. These effects are based on the ability of CAS proteins to act as scaffolds for other catalytically active or adaptor proteins.

In their initial report of EFS, Ishino et al. (1995) showed the protein bound directly to the SRC-family kinase Fyn in vitro and in vivo, and showed that Fyn directly phosphorylated EFS. Alexandropoulos and colleagues showed that SRC directly phosphorylates residues Y576 and Y577 tyrosine sites on the EFS, thus enhancing FAK targeting, and eventually the solubility and/or stability of the complex. Reciprocally, EFS activates SRC signaling through c-CRK and RAP1 (Xing et al., 2000), a process associated with cell migration for multiple family members (Fashena et al., 2002; Cary et al., 1998). EFS contains two high-affinity class I Src-SH3-binding sites (Alexandropoulos et al., 1995); it might be involved in the regulation of c-Src. Through a series of in vitro and in vivo assays with truncated and full length EFS constructs, this group carefully dissected the mechanism of c-SRC–EFS complex formation and further activation of c-SRC by EFS (Alexandropoulos and Baltimore, 1996), as necessary for allowing a direct interaction between EFS and CRK. This work defined the DVP motifs carboxy-terminal to the phosphorylated tyrosines in the EFS substrate domain as contributing to a consensus recognition site for the CRK SH2 domain (Songyang et al., 1993).

Activation of EFS to reorganize the actin cytoskeleton downstream of CRK has been shown to play a role in neurite outgrowth. Yang et al. identified EFS as an essential component of an additional signaling pathway involved in neurite processes formation, connected to the canonical ERK activation pathway (Stofega et al., 1997; Hakak and Martin, 1999). RPTP α (receptor–protein–tyrosine phosphatase α) is abundant in neural tissue and associates physically with SRC and Fyn (Zheng et al., 2000; Bhandari et al., 1998), and its overexpression activates these kinases

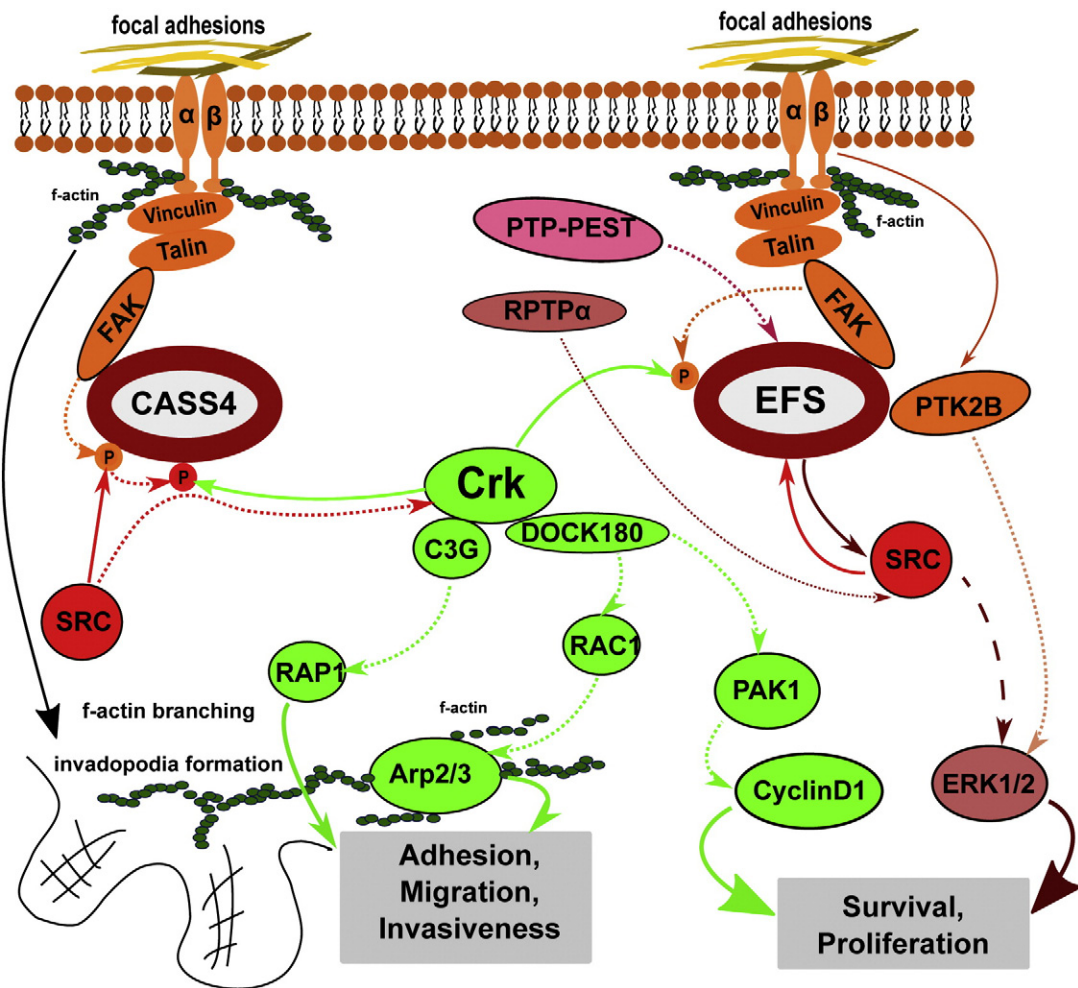


Fig. 2. Scheme of signaling pathways involving EFS and CASS4 proteins. Integrin ($\alpha\beta$) heterodimer activation at focal adhesions triggers phosphorylation of CAS proteins by FAK and SRC, allowing them to act as scaffolds for CRK and downstream factors that reorganize the actin cytoskeleton, and influence proliferation and survival signaling. Dotted lines stand for phosphorylation events, solid lines stand for direct interactions between proteins.

(den Hertog et al., 1993) in response to EGF stimulation. In this study, the authors showed that stimulation by RPTP α caused SRC phosphorylation of EFS, and increased recruitment of CRK and NCK to EFS (Yang et al., 2002). Alexandropoulos et al. have also investigated EFS expression in T lymphocytes and in immune responses (Alexandropoulos et al., 2003), discussed further below.

Very few papers address the mechanistic action of CASS4, with these beginning with the hypothesis that it has biological activities similar to those of other family members in influencing cell attachment and movement. Singh et al. (2008) as stated above, confirmed direct interaction between CASS4 SH3 domain and FAK, also authors revealed colocalization of CASS4 with paxillin within focal adhesions in HOP-62 cells. In direct comparison to NEDD9, overexpression of CASS4 was shown to act similarly in activating FAK phosphorylation and promoting cell spreading, but to a lesser degree than for NEDD9, and only in a subset of cells overexpressing the protein. Depletion of CASS4 reduced the spreading and FAK activation in a subset of cells. Despite the fact that CASS4 lacks the YDYVHL site that coordinates docking with the Src SH2 domain for other family members (Tachibana et al., 1997), attachment-induced Src phosphorylation of CASS4 was observed, suggesting that additional interactions between CASS4, FAK, and Src are sufficient to drive this modification. Unusually, CASS4 depletion had a bimodal effect on cell migration, causing some cells to have lower velocity and others to have higher velocity than control cells. This study

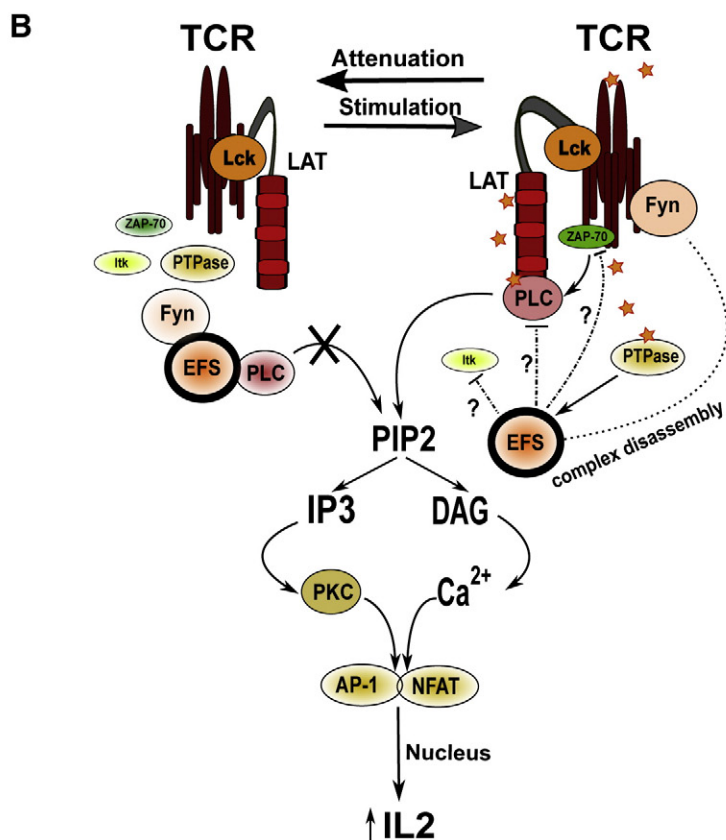
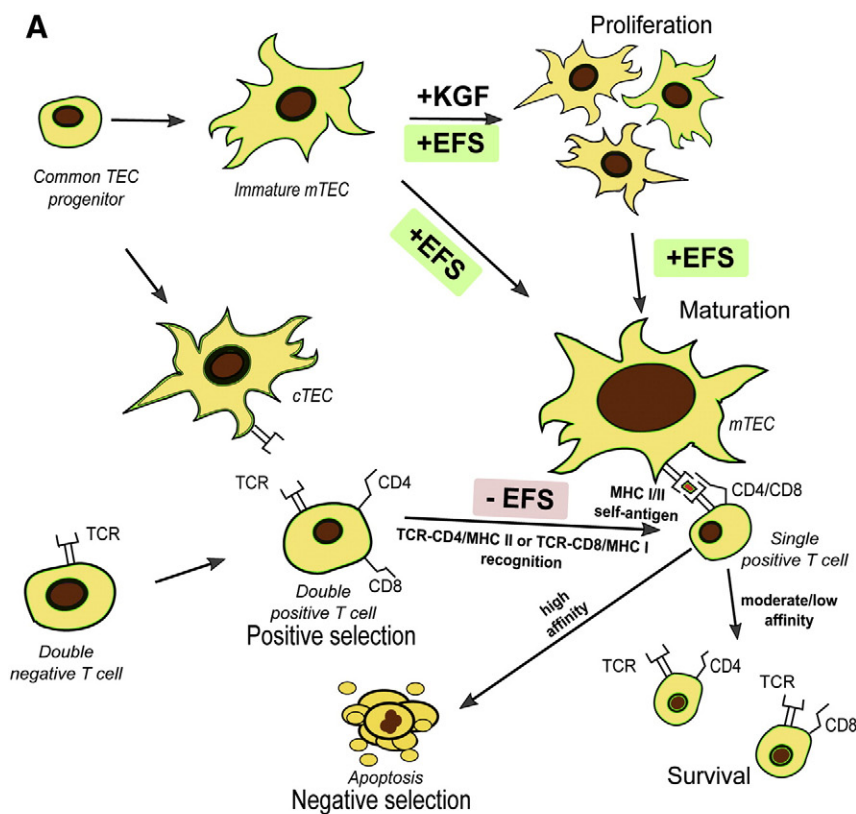
concluded that CASS4 was a weaker paralogue of BCAR1, and also suggested that the function of CASS4 may be cell-type specific and dependent upon the presence or absence of expression of other CAS family members because of the structural differences noted above.

6. EFS and immune system function

Several studies have addressed the role of EFS in T-cell function and maturation, indicating important roles in maintaining immune homeostasis and preventing development of autoimmunity (Fig. 3A). Medullary thymus epithelial cells (mTECs) are important for proper T-cell maturation and negative selection of autoreactive clones, required for development of central immunological self-tolerance. Two studies have reported that EFS expression by (mTECs) is important for negative selection of T-cells during their development, while cortical thymus epithelial cells, required for positive T cell selection, were shown not to express EFS at the protein level (Donlin et al., 2005; Danzl et al., 2010). EFS expression in mTECs was shown to be mechanistically crucial for their functional maturation and keratinocyte growth-factor (KGF)-induced expansion. In these studies, mice with genetic reduction of EFS levels developed normally during embryogenesis but after 7–14 months of life developed massive inflammatory lesions with T-cell infiltration in multiple tissues including small intestine, liver, kidneys and lungs that bore a striking histological resemblance to inflammatory bowel

disorders similar to Crohn's disease (Donlin et al., 2005). In this context, it is interesting that (He et al., 2013) linked EFS single nucleotide polymorphisms (SNPs) to Crohn's disease using a novel Sherlock algorithm

on a study of 3230 cases versus 4829 controls despite the fact that initial genome-wide association studies (GWAS) of Crohn's disease did not identify EFS (Barrett et al., 2008). Then they performed a replication



analysis on an independent sample from another GWAS (Franke et al., 2010) of total 6333 patients with Crohn's disease versus 15,056 controls and confirmed this association (p-value 0.039). The 16 SNPs linked to EFS in this work were trans-acting, potentially affecting the level of EFS expression but not its coding sequence.

Besides a role in mTECs, there is also evidence that EFS-mediated signaling in lymphocytes is crucial during their maturation. Donlin et al. (2002) have shown that transgenic expression of a truncated form of EFS (SinDeltaC) under the human CD2 promoter, which causes transgene expression in both thymocytes and mature T cells, resulted in a constitutively activated protein that was bound more effectively and was hyperphosphorylated by the SRC-family Fyn kinase. Lymphocyte maturation was impaired beyond the double positive stage (CD4+ CD8+), with the transgenic phenotype associated with 2-fold increased lymphocyte apoptosis and decreased levels of single positive (CD4+ CD8– and CD4– CD8+) T cells. Crossing these mice to Fyn heterozygous or null mice alleviated the apoptotic phenotype, but mice still were deficient in development of single positive cells, suggesting that additional SRC kinases or other kinases may be involved in this process. SinDeltaC expression did not affect T cell receptor (TCR)-proximal signaling or JNK activity, but did inhibit the ERK activation that is required during lymphocytes positive selection (Alberola-Ila et al., 1996). These results indicated that EFS inactivation (dephosphorylation) was necessary for appropriate T cell differentiation.

In other work, EFS was shown to play repressive role on processes associated with the activation of mature T-cells, including IL-2 pro-inflammatory cytokine secretion and IL-2-dependent clonal expansion of T cells (Donlin et al., 2005; Xing et al., 2004). In contrast to many other signal transducers which are phosphorylated upon TCR stimulation, EFS was found to be phosphorylated and bound to Fyn kinase and phospholipase C- γ in resting T-cells. EFS dephosphorylation after T-cell receptor (TCR) stimulation and release of the SRC family kinase Fyn and phospholipase C- γ normally lead to self-limitation of the immune response. EFS overexpression in murine wild type T cells decreased IL-2 concentration in supernatants versus controls in response to TCR stimulation (Xing et al., 2004). Conversely, in Jurkat T cells with EFS expression reduced by siRNA, NFAT mediated transcriptional activation was increased (Donlin et al., 2005). The authors of one study (Xing et al., 2004) have proposed a complex dualistic role of EFS in mature T cells functions, because both overexpression and siRNA knockdown of this protein in cell models resulted in decreased transcriptional activation of IL-2 dependent promoters following TCR stimulation (Fig. 3b).

Another study has raised the possibility that elevated EFS expression might contribute to susceptibility to acute rheumatoid fever (Bryant et al., 2014). In this work, peripheral blood mononuclear cells (PBMCs) from patients with rheumatoid heart disease (RHD) and control subjects that had never experienced acute rheumatoid fever were stimulated with rheumatogenic and non-rheumatogenic group A streptococci (GAS) strains. EFS is one of the only four genes with significantly increased expression associated with rheumatogenic GAS exposure. In a possibly related observation, Tchernev et al. (2002) have connected EFS function to Chediak–Higashi syndrome (CHS) which is rare and severe

autosomal recessive disorder associated with partial albinism, peripheral neuropathy, mild coagulation defects and propensity to recurrent bacterial and fungal infections, caused by incomplete phagocytosis due to failure in phagolysosome formation. In this study, the authors identified a direct physical interaction in vitro and in vivo between EFS and LYST (lysosomal trafficking regulator, aka CHS1 (Chediak–Higashi syndrome 1)), a large protein that is mutated in CHS and regulates intracellular trafficking of proteins through endosomes. These results may imply the role of EFS as a disease progression modifier, although further testing and establishment of mechanism is necessary. The connections to CHS and to GAS infection is also of interest because of the well-documented role of BCAR1 in regulating infection by microbial pathogens including Salmonella, Plasmodium, herpesvirus, and Yersinia noted above (Bruce-Staskal et al., 2002; Weidow et al., 2000), and recent implication of NEDD9 as a regulator of infection by *Candida albicans* (Liu et al., 2015): it is possible that EFS has similar function. This in turn may be relevant to a role in Crohn's disease, which is linked to recurrent infection with pathogenic microorganisms (Stappenbeck et al., 2011). In sum, these data are consistent with the idea that EFS makes multiple contributions to T lymphocyte development and immune system maturation and that it acts both directly in lymphocytes and indirectly through mTECs.

By contrast, almost no work has addressed possible roles of CASS4 in immune system function. However, changes in CASS4 expression have been linked to atopic asthma in a study performed by Esnault et al. (2013). In this work, CASS4 was reported to be an eosinophil-associated gene, with expression in sputum cells significantly increased after whole lung allergen challenge. The CASS4 mRNA was also upregulated in cells collected by bronchoalveolar lavage after segmental broncho-provocation with an allergen. When this procedure was performed following administration of mepolizumab (a humanized monoclonal anti-IL-5 antibodies which reduces excessive eosinophilia) CASS4 mRNA was reciprocally downregulated, therefore suggesting relation of CASS4 activity with immune response during atopic asthma development. Direct functional testing is warranted.

7. Implication of EFS and CASS4 in other disorders

A number of studies not directly focused on EFS or CASS4 have provided initial suggestions that the activity of these proteins is relevant to other diseases (summarized in Table 2).

7.1. Alzheimer's disease

Alzheimer's disease has been shown to have strong pathological association with CASS4 by the IGAP (International Genomics of Alzheimer's Disease Project), the largest genetic epidemiological investigation of AD risk to date, involving assessment of 74,046 patients. In this work, Lambert et al. (2013) found a locus associated with lower susceptibility to AD on chromosome 20 at the location of the CASS4 gene, and identified a corresponding SNP – rs7274581 T/C (OR = 0.87) – which reached genome-wide significance in both the first stage (p-value 1.6×10^{-6} in the analysis of 4 combined samples totaling 17,008

Fig. 3. EFS and immune system activity. A. A common thymic epithelial cell (TEC) progenitor gives rise to cortical TECs (cTECs) and medullary TECs (mTECs). Immature mTECs are distinguished by lack of expression of the Aire (autoimmune regulator) transcription factor. Mature TECs have high expression of the Major Histocompatibility Complexes (MHC I and MHC II), which are required for representation of antigens, including self-antigens. During their differentiation, T lymphocytes progress from a double negative (CD4– CD8–) to a double positive (CD4+ CD8+) stage, at which point positive selection by cTECs occurs. Only T cells moderately binding to the MHC I (for CD8+) or MHC II (for CD4+) molecules expressed by cTECs are provided with a pro-survival co-stimulatory signal. EFS dephosphorylation and downregulation of its effectors in T lymphocytes is crucial for them to move from the double positive to the single positive (CD4+ CD8– or CD4– CD8+) stage. This process is accompanied by T cell migration to the thymic medulla, where negative selection by mTECs to remove auto-reactive clones occurs. EFS signaling is required for KGF (keratinocyte growth factor)-mediated expansion and functional maturation of mTECs. T cells binding MHC–self-antigen complexes with high affinity undergo apoptosis. B. A dual role of EFS in mature T cells was identified in a Jurkat T cell model (Xing et al., 2004). Both overexpression and siRNA knockdown of EFS led to decreased transcriptional activation of IL-2-dependent promoters following TCR stimulation. Phosphorylated EFS bound to Fyn and PLC- γ sequesters signaling molecules away from the TCR complex, acting as a negative regulator in resting cells. TCR activation leads to dephosphorylation of EFS, most likely through the action of a protein tyrosine phosphatase (PTPase), resulting in release of binding partners and activating downstream signaling: for instance, activated PLC- γ cleaves phosphoinositide PIP2 into the second messengers IP3 and DAG. IP3 induces intracellular calcium release, causing activation of NFAT, and DAG activates PKC, resulting in increased AP-1 transcriptional activity: NFAT and AP-1 bind and contribute to activation of the IL2 promoter. Attenuation of TCR signaling induces inhibitory reassociation of the EFS–Fyn–PLC γ complex.

Table 2

Identification of EFS or CASS4 in studies employing high throughput or GWAS screening methods.

Disease associations			
Screen purpose	Protein	Observation	Reference
<i>GWAS</i>			
Alzheimer's disease	CASS4	SNP rs7274581 T/C linked to risk. Odds ratio 0.72; p-value 0.011 SNP rs7274581 T/C linked to risk. Odds ratio 0.88; p-value 2.5×10^{-8} SNP rs7274581 T/C linked to risk. Odds ratio 0.8888; p-value 1.75×10^{-7} SNP rs6024870, RegulomeDB score 2b, meaning that this SNP is likely to affect transcription factor binding. SNP rs16979934 T/G linked to risk. Odds ratio 0.5956; p-value 0.03.	Beecham et al. (2014) Lambert et al. (2013) Ruiz et al. (2014) Rosenthal et al. (2014)
Cystic fibrosis	CASS4	Possible correlation with severity of the lung manifestation of the disease	Wang et al. (2015)
Crohn's disease	EFS	EFS gene linkage to Crohn's disease confirmation (p-value 0.039) in humans	Wright et al. (2011) He et al. (2013)
<i>Expression or gene dosage change association</i>			
Atopic asthma	CASS4	Significantly increased expression in eosinophils after antigen exposure	Esnaul et al. (2013)
Non-small cell lung cancer	CASS4	Association of high CASS4 expression in tumors with disease severity and poor prognosis	Miao et al. (2013)
Colorectal cancer	CASS4	Increased CASS4 expression is associated with higher rate of chromosomal instabilities and worse prognosis in colorectal cancer.	Bond et al. (2014)
Rheumatic fever	EFS	Significant increase in expression after peripheral blood mononuclear cells stimulation with rheumatogenic group A streptococci from patients with rheumatoid heart disease	Bryant et al. (2014)
Prostate cancer	EFS	EFS gene CpG sites hypermethylation and decreased EFS expression are associated with prostate cancer recurrence and metastatic potential.	Sertkaya et al. (2015)
Uveal melanoma	EFS	EFS hypermethylation is associated with high metastatic rate.	Neumann et al. (2011)
Breast cancer	EFS	EFS expression may play role in trastuzumab resistance development in HER2+ breast cancer.	Boyer et al. (2013)
Prolactinoma	EFS	EFS association to stem cell regulation, tumor cell invasion, tumor recurrence, and drug resistance	Tong et al. (2012)
Gestational choriocarcinoma	EFS	EFS is one of more than 100 genes located in frequently amplified chromosomal region.	Poaty et al. (2012)
Glioblastoma multiforme	EFS	Differential expression of EFS between glioblastoma multiforme subtypes	Vital et al. (2010)
Endometrium expression profiling	EFS	17 β -Estradiol or progesterone stimulation decreases EFS expression in explants of late proliferative phase endometrium	Dassen et al. (2007)
<i>Other</i>			
Phosphoproteome of resting human platelets	CASS4	Phosphorylation on S305 PKA/PKG consensus site	Zahedi et al. (2008)
Platelet activation by oxidized phospholipids	CASS4	KODA-PC induces S249 phosphorylation in platelets	Zimman et al. (2014)
Chediak-Higashi syndrome	EFS	Direct interaction with LYST	Tchernev et al. (2002)

cases versus 37,154 controls) and the second stage (p-value 4.1×10^{-3} in the replication study by genotyping SNPs showed moderate significance in stage 1 in an independent sample of 8572 cases versus 11,312 controls) analysis. Several additional studies have investigated this association. Ruiz et al. conducted a follow-up study comparing 1808 patients with AD symptoms versus 2564 unrelated healthy individuals (Ruiz et al., 2014), with results showing that this SNP was not predictive. However, in the GWAS performed by Beecham et al. (2014) CASS4 showed a significant correlation with clinical pathological features of AD such as neurofibrillary tangles and neuritic plaques. Two additional CASS4 SNPs were reported to be associated with AD susceptibility: rs6024870 (RegulomeDB score 2b) by Rosenthal et al. (2014) and rs16979934 T/G (OR = 0.5956, p-value 0.03) by Wang et al. (2015). Given the likely conserved CAS family cytoskeletal function of CASS4, summarized above, it has been speculated that it may have a role in axonal transport and influence the expression of the amyloid precursor (APP) and tau proteins, which are pathologically affected in AD (Karch and Goate, 2015). Possible mechanisms for CASS4 action in AD are discussed by Beck et al. (2014).

7.2. Cystic fibrosis

Cystic fibrosis severity, progression and comorbid conditions have been reported to be associated with altered expression of CASS4. A GWAS study performed by Wright et al. (2011) involved 3467 individuals in three independent cohorts with different recruitment parameters. The Gene Modifier Study (GMS) included unrelated affected individuals homozygous for CTFR allele p.Phe508del, common in cystic fibrosis; the Canadian Consortium for Genetic Studies (CGS) was a population-based study of patients with cystic fibrosis; and the Twins & Sibs Study (TSS) assessed families with two or more surviving

children with cystic fibrosis. Evidence of CASS4 association was first identified in patients in the GMS and CGS groups, and then linkage was analyzed in 486 sibling pairs in family-based TSS. A number of modifier loci for lung disease severity in cystic fibrosis were identified during this analysis, including the 20q13.2 locus ($\log_{10}$ odds = 5.03) that contains five genes (CBLN4, MC3R, CASS4, CSTF1, and AURKA), expressed in fetal or adult lung, or bronchial epithelia. Taking in consideration that the CAS family member NEDD9 has been shown to interact directly with AURKA (encoding Aurora-A kinase) in cell cycle regulation (Pugacheva and Golemis, 2005) and ciliary resorption (Pugacheva et al., 2007), it is possible that CASS4 may similarly interact with Aurora-A kinase. Further investigation is necessary.

7.3. Thrombosis

CASS4 signaling may contribute to platelet activation and aggregation. Zahedi et al. (2008) performed a phosphoproteome analysis of resting human platelets and identified a PKA/PKG phosphorylation site in CASS4 on residue S305 in the substrate domain; the functional significance of this phosphorylation is currently unknown. Zimman et al. (2014) showed significantly increased phosphorylation on S249 of CASS4, also in the substrate domain, after platelet stimulation with the oxidized phospholipid KODA-PC (9-keto-12-oxo-10-dodecanoic acid ester of 2-lyso-phosphocholine, a CD36 receptor agonist) versus thrombin treatment, which may implicate CASS4-mediated signaling in platelet hyperreactivity. Direct evaluation of these ideas is required.

7.4. Cancer

Many CAS family proteins have altered activity and functional roles in cancer initiation, progression, metastasis, and drug resistance (Nikonova

et al., 2014b; Tornillo et al., 2014). Based on information in the genomic databases available through the cBioportal site (The Cancer Genome Atlas (TCGA) and others), the CASS4 and EFS genes show copy number, gene expression changes, or mutation in a subset of tumors for most tumor types for which information is available (Fig. 4). CASS4 is more frequently amplified than EFS, particularly in tumor types that often have genomic instability and a higher mutational burden: this may reflect a specific preference for elevated CASS4, or alternatively may reflect the fact that CASS4 is part of a common amplicon with the AURKA oncogene, which is often elevated in cancer (and see below discussion). Gene expression changes generally indicate the CAS genes, and particularly EFS, are more commonly overexpressed than downregulated in tumors. More thorough analysis may assign changes in CAS expression to tumor stage and grade that provide insights into roles of these proteins at specific points in tumor progression.

In focused analyses, a number of studies connect EFS to prostate cancer. In one study, the local and systemic recurrence of prostate cancer was associated with CpG site hypermethylation of number of genes, including EFS, FLNC, ECRG4, PITX2, PDLIM4, and KCNMA1. Further investigation indicated that methylation of CpG sites that is specifically predicted to result in reduction of gene expression was restricted to the FLNC and EFS genes ($p \leq .03$), both involved in cell attachment (Vanaja et al., 2009). In a study conducted by Marques et al., EFS expression was strongly downregulated in hormonal therapy-resistant versus therapy-responsive PC346C prostate cancer cells (Marques et al., 2010), in general agreement with a second study, in which low EFS expression correlated with malignant behavior of the PC-3 and LNCaP prostate cancer cells (Sertkaya et al., 2015). Additional work showed decreased EFS

mRNA expression levels in prostate cancer samples with higher Gleason scores (Nakagawa et al., 2008). Finally, in a 2013 study of castration-resistant prostate cancer, EFS was identified as having significantly increased gross phosphorylation levels in samples from short or long term androgen-deprived (AD), or castration-resistant prostate carcinoma xenografts, versus in AD-naïve xenograft samples (Roe et al., 2013). This paints a complex picture of reduced expression, but increased protein activity, in the case of aggressive disease, and requires further study.

Methylation of the EFS CpG island was observed in 69% of cases of uveal melanoma (UM) (Neumann et al., 2011), with methylation only observed in cases of metastatic disease. As anticipated, RT-PCR expression analysis revealed a significant inverse relationship between EFS mRNA expression with EFS methylation in UM. EFS methylation was tissue-specific with full methylation in peripheral blood cells, but no methylation in other tissues such as fetal muscle, kidney and brain was detected.

EFS and other proteins involved in SRC kinase signaling (CDCP1/Trask and Paxillin) were showed to have increased expression in a study of breast cancer using trastuzumab (Herceptin)-resistant versus -sensitive BT474 cell line derivatives (Boyer et al., 2013). Importantly, EFS knockdown with siRNA restored trastuzumab sensitivity (Boyer et al., 2013). Reflecting the importance of post-translational modification of CAS proteins, in a study of cell lines and tumor tissue in malignant melanoma, EFS phosphorylation and activity significantly decreased ($p < 0.05$) in response to vemurafenib treatment in *BRAF* wild-type melanoma tumors compared to tumors with a *BRAF* V600E mutation with additional resistance to vemurafenib (Tahiri et al., 2013).

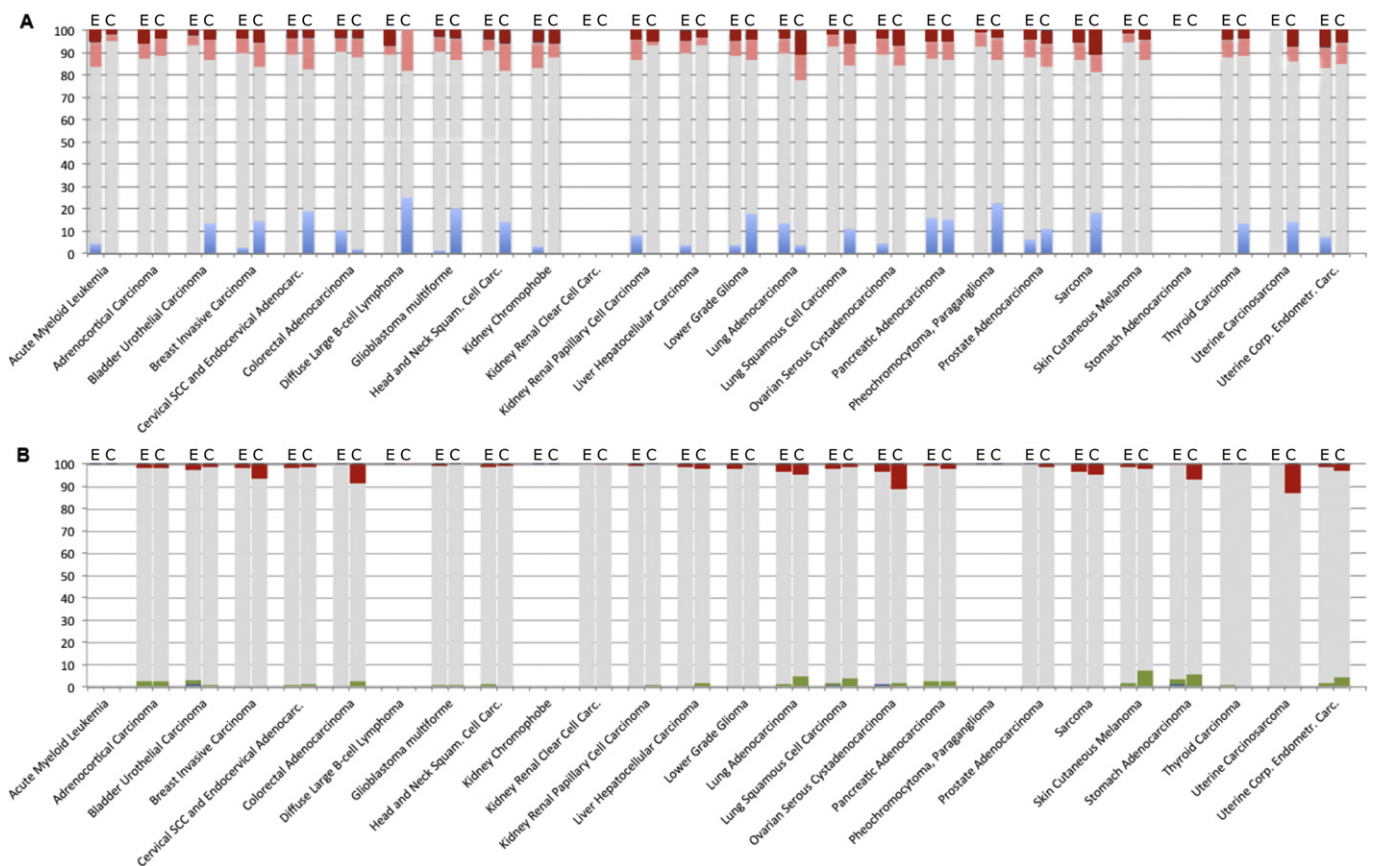


Fig. 4. Genomic alterations and expression changes in CASS4 and EFS in cancer. Data available from the TCGA was extracted using the cBioPortal site (<http://www.cbioportal.org>). For some tumor sites, data was unavailable for both genomic and mRNA profiles, reflected by empty columns. A. mRNA expression profiles for EFS (E) and CASS4 (C). Dark red: percent of tumor samples with mRNA upregulation with Z-score > 2; pink: upregulation with Z-score > 1; light blue: percent of tumor samples with mRNA downregulation Z-score < -1; gray: no significant change ($-1 < z < 1$). B. Copy number variations and somatic mutations for EFS (E) and CASS4 (C). Red: percent of tumor samples with gene amplifications; blue: homozygous deletions; green: mutations. Gray represents no change. The fraction of samples with both CNV and mutations in the genes of interest was not significant and is not shown.

In more tenuous connections to cancer, gestational choriocarcinoma is characterized by overexpression of the centromeric 10.21 Mb “minimal critical region” on Chromosome 14, containing more than 100 genes, including EFS (Poaty et al., 2012). The EFS mRNA was also identified as differentially expressed in two of the three groups of glioblastoma multiforme as identified by gene expression profiles (GEPs) (Vital et al., 2010). EFS was differentially expressed in GEP1 and GEP3 groups, which were associated with worse prognosis, and increased cytogenetic abnormalities.

CASS4 has been much less studied in the context of human malignancies. Miao et al. correlated elevated CASS4 expression with lymph node metastasis and high TNM stage for non-small cell lung cancer (NSCLC) (Miao et al., 2013). In addition, this study detected a significant difference in cytoplasmic accumulation of CASS4 protein between high (H1299 and BE1) and low (LTE and A549) metastatic potential lung cancer cell lines. These may suggest CASS4 as a possible prognostic marker in clinical management of NSCLC. CASS4 was highlighted in a study by Bond et al. (2014), which performed a stratification of the BRAF (V600E) microsatellite stable (MSS) and microsatellite-unstable (MSI) mutations in colorectal cancers. MSS mutations confer a poor patient prognosis, whereas MSI mutant colorectal cancers have an excellent prognosis. CASS4 was among the genes located on chromosome 20q (along with AURKA, BCAS1 and ZBP1 genes) that displayed statistically significant ($p = 0.01$) increase in the presence of copy number aberrations that are indicative of chromosomal instability in BRAF mutant/MSS cancers ($n = 33$) compared to BRAF mutant/MSI cancers ($n = 30$). This study suggested CASS4 and associated genes possible prognostic markers in human colorectal pathologies.

8. Conclusions

Twenty years after its initial description, EFS is now emerging as an important determinant of multiple immune cell processes relevant to auto-immunity and control of infection. EFS deserves more attention as a potential biomarker for these conditions, and study of EFS is likely to inform understanding of disease mechanisms for conditions including Crohn's disease, CHS, and other inflammatory syndromes. For the more recently identified CASS4, the most intriguing direct linkage to a clinical condition is to Alzheimer's disease, where it may prove of value as a predictive biomarker, and study of CASS4 in this setting may help elucidate underlying pathogenic processes. For both proteins, expression or phosphorylation may be useful as a marker of disease progression and prognosis in some types of cancer. There are currently no therapeutic approaches targeting EFS and CASS4, and given the proteins lack a catalytic domain and extracellular moieties, it may be challenging to generate such agents. However, given increasingly detailed knowledge of conserved versus unique domain-specific interactions of EFS, CASS4, and their partner proteins, it is possible that protein–interaction disrupting tools can be devised. Clearly, greater investigation of CASS4 and EFS in their own right, rather than as less commonly expressed CAS paralogues, is now entirely merited.

Acknowledgments

We thank Dr. Ilya Serebriiskii for guidance in analyzing data in cBioPortal. The authors were supported by NCI Core Grant P30 CA006927 (to Fox Chase Cancer Center) and NCI CA181287 (to EAG); by a project of the Russian Government to support the Program for Competitive Growth of Kazan Federal University (to AD); and by the Russian Ministry of Science and Education, as task #17.1891.2014/K (to VK). This review and two corresponding Gene Wiki articles were written as part of the Gene Wiki Review series—a series resulting from a collaboration between the journal GENE and the Gene Wiki Initiative. The Gene Wiki Initiative is supported by the National Institutes of Health (GM089820). Additional support for Gene Wiki Reviews is provided by Elsevier, the publisher of GENE. The corresponding Gene

Wiki entries for this review can be found here: http://en.wikipedia.org/wiki/Embryonal_fyn-associated_substrate and <http://en.wikipedia.org/wiki/CASS4>.

References

- Alberola-Illa, J., et al., 1996. Positive and negative selection invoke distinct signaling pathways. *J. Exp. Med.* 184 (1), 9–18.
- Alexandropoulos, K., Baltimore, D., 1996. Coordinate activation of c-Src by SH3- and SH2-binding sites on a novel p130Cas-related protein. *Sin. Genes Dev.* 10 (11), 1341–1355.
- Alexandropoulos, K., Cheng, G., Baltimore, D., 1995. Proline-rich sequences that bind to Src homology 3 domains with individual specificities. *Proc. Natl. Acad. Sci. U. S. A.* 92 (8), 3110–3114.
- Alexandropoulos, K., et al., 2003. Sin: good or bad? A T lymphocyte perspective. *Immunol. Rev.* 192, 181–195.
- Aquino, J.B., et al., 2009. The retinoic acid inducible Cas-family signaling protein Nedd9 regulates neural crest cell migration by modulating adhesion and actin dynamics. *Neuroscience* 162 (4), 1106–1119.
- Astier, A., et al., 1997. The related adhesion focal tyrosine kinase differentially phosphorylates p130Cas and the Cas-like protein, p105HEF1. *J. Biol. Chem.* 272 (32), 19719–19724.
- Bandayopadhyay, C., et al., 2014. p130Cas scaffolds the signalosome to direct adaptor-effector cross talk during Kaposi's sarcoma-associated herpesvirus trafficking in human microvascular dermal endothelial cells. *J. Virol.* 88 (23), 13858–13878.
- Barrett, J.C., et al., 2008. Genome-wide association defines more than 30 distinct susceptibility loci for Crohn's disease. *Nat. Genet.* 40 (8), 955–962.
- Barrett, A., et al., 2013. p130Cas: a key signalling node in health and disease. *Cell. Signal.* 25 (4), 766–777.
- Beck, T.N., et al., 2014. Adaptors for disorders of the brain? The cancer signaling proteins NEDD9, CASS4, and PTK2B in Alzheimer's disease. *Oncoscience* 1 (7), 486–503.
- Beecham, G.W., et al., 2014. Genome-wide association meta-analysis of neuropathologic features of Alzheimer's disease and related dementias. *PLoS Genet.* 10 (9), e1004606.
- Bhandari, V., Lim, K.L., Pallen, C.J., 1998. Physical and functional interactions between receptor-like protein-tyrosine phosphatase alpha and p59fyn. *J. Biol. Chem.* 273 (15), 8691–8698.
- Bond, C.E., et al., 2014. Microsatellite stable colorectal cancers stratified by the BRAF V600E mutation show distinct patterns of chromosomal instability. *PLoS One* 9 (3).
- Boyer, A.P., et al., 2013. Quantitative proteomics with siRNA screening identifies novel mechanisms of trastuzumab resistance in HER2 amplified breast cancers. *Mol. Cell. Proteomics* 12 (1), 180–193.
- Brehme, M., et al., 2009. Charting the molecular network of the drug target Bcr-Abl. *Proc. Natl. Acad. Sci. U. S. A.* 106 (18), 7414–7419.
- Bruce-Staskal, P.J., et al., 2002. Cas, Fak and Pyk2 function in diverse signaling cascades to promote Yersinia uptake. *J. Cell Sci.* 115 (Pt 13), 2689–2700.
- Bryant, P.A., et al., 2014. Susceptibility to acute rheumatic fever based on differential expression of genes involved in cytotoxicity, chemotaxis, and apoptosis. *Infect. Immun.* 82 (2), 753–761.
- Cary, L.A., et al., 1998. Identification of p130Cas as a mediator of focal adhesion kinase-promoted cell migration. *J. Cell Biol.* 140 (1), 211–221.
- Cote, J.F., et al., 1998. Combination of gene targeting and substrate trapping to identify substrates of protein tyrosine phosphatases using PTP-PEST as a model. *Biochemistry* 37 (38), 13128–13137.
- Danzl, N.M., Donlin, L.T., Alexandropoulos, K., 2010. Regulation of medullary thymic epithelial cell differentiation and function by the signaling protein Sin. *J. Exp. Med.* 207 (5), 999–1013.
- Dassen, H., et al., 2007. Progesterone regulation of implantation-related genes: new insights into the role of oestrogen. *Cell. Mol. Life Sci.* 64 (7–8), 1009–1032.
- Davidson, D., Veilleux, A., 2001. PTP-PEST, a scaffold protein tyrosine phosphatase, negatively regulates lymphocyte activation by targeting a unique set of substrates. *EMBO J.* 20 (13), 3414–3426.
- Davis, S.P., et al., 2012. *Plasmodium falciparum*-induced CD36 clustering rapidly strengthens cytoadherence via p130CAS-mediated actin cytoskeletal rearrangement. *FASEB J.* 26 (3), 1119–1130.
- den Hertog, J., et al., 1993. Receptor protein tyrosine phosphatase alpha activates pp60c-src and is involved in neuronal differentiation. *EMBO J.* 12 (10), 3789–3798.
- Donlin, L.T., et al., 2002. Defective thymocyte maturation by transgenic expression of a truncated form of the T lymphocyte adapter molecule and Fyn substrate, Sin. *J. Immunol.* 169 (12), 6900–6909.
- Donlin, L.T., et al., 2005. Deficiency in expression of the signaling protein Sin/Efs leads to T-lymphocyte activation and mucosal inflammation. *Mol. Cell. Biol.* 25 (24), 11035–11046.
- EntrezGene, 2015a. EFS gene. <http://www.ncbi.nlm.nih.gov/gene/10278>.
- EntrezGene, 2015b. CASS4 gene. <http://www.ncbi.nlm.nih.gov/gene/57091>.
- Esnaul, S., et al., 2013. Identification of genes expressed by human airway eosinophils after an in vivo allergen challenge. *PLoS One* 8 (7), e67560.
- Fashena, S.J., et al., 2002. Dissection of HEF1-dependent functions in motility and transcriptional regulation. *J. Cell Sci.* 115 (Pt 1), 99–111.
- Franke, A., et al., 2010. Genome-wide meta-analysis increases to 71 the number of confirmed Crohn's disease susceptibility loci. *Nat. Genet.* 42 (12), 1118–1125.
- Garton, A.J., et al., 1997. Association of PTP-PEST with the SH3 domain of p130Cas; a novel mechanism of protein tyrosine phosphatase substrate recognition. *Oncogene* 15 (8), 877–885.

- Gonsior, C., et al., 2014. Oligodendroglial p130Cas is a target of Fyn kinase involved in process formation, cell migration and survival. *PLoS One* 9 (2), e89423.
- Hakak, Y., Martin, G.S., 1999. Cas mediates transcriptional activation of the serum response element by Src. *Mol. Cell. Biol.* 19 (10), 6953–6962.
- Harte, M.T., et al., 1996. p130Cas, a substrate associated with v-Src and v-Crk, localizes to focal adhesions and binds to focal adhesion kinase. *J. Biol. Chem.* 271 (23), 13649–13655.
- He, X., et al., 2013. Sherlock: detecting gene-disease associations by matching patterns of expression QTL and GWAS. *Am. J. Hum. Genet.* 92 (5), 667–680.
- Hsia, D.A., et al., 2003. Differential regulation of cell motility and invasion by FAK. *J. Cell Biol.* 160 (5), 753–767.
- Ishino, M., et al., 1995. Molecular cloning of a cDNA encoding a phosphoprotein, Efs, which contains a Src homology 3 domain and associates with Fyn. *Oncogene* 11 (11), 2331–2338.
- Ishino, M., et al., 1997. Identification of an Efs isoform that lacks the SH3 domain and chromosomal mapping of human Efs. *Oncogene* 15 (14), 1741–1745.
- Karch, C.M., Goate, A.M., 2015. Alzheimer's disease risk genes and mechanisms of disease pathogenesis. *Biol. Psychiatry* 77 (1), 43–51.
- Kirsch, K.H., Georgescu, M.M., Hanafusa, H., 1998. Direct binding of p130(Cas) to the guanine nucleotide exchange factor C3G. *J. Biol. Chem.* 273 (40), 25673–25679.
- Lambert, J.C., et al., 2013. Meta-analysis of 74,046 individuals identifies 11 new susceptibility loci for Alzheimer's disease. *Nat. Genet.* 45 (12), 1452–1458.
- Law, S.F., et al., 1996. Human enhancer of filamentation 1, a novel p130cas-like docking protein, associates with focal adhesion kinase and induces pseudohyphal growth in *Saccharomyces cerevisiae*. *Mol. Cell. Biol.* 16 (7), 3327–3337.
- Law, S.F., et al., 1999. Dimerization of the docking/adaptor protein HEF1 via a carboxy-terminal helix–loop–helix domain. *Exp. Cell Res.* 252 (1), 224–235.
- Liu, F., Sells, M.A., Chemoff, J., 1998. Protein tyrosine phosphatase 1B negatively regulates integrin signaling. *Curr. Biol.* 8 (3), 173–176.
- Liu, Y., et al., 2015. New signaling pathways govern the host response to *C. albicans* infection in various niches. *Genome Res.* 25 (5), 679–689.
- Marques, R.B., et al., 2010. Bypass mechanisms of the androgen receptor pathway in therapy-resistant prostate cancer cell models. *PLoS One* 5 (10), e13500.
- Miao, Y., et al., 2013. Overexpression and cytoplasmic accumulation of Hepl1 is associated with clinicopathological parameters and poor prognosis in non-small cell lung cancer. *Tumour Biol.* 34 (1), 107–114.
- Minegishi, M., et al., 1996. Structure and function of Cas-L, a 105-kD Crk-associated substrate-related protein that is involved in beta 1 integrin-mediated signaling in lymphocytes. *J. Exp. Med.* 184 (4), 1365–1375.
- Nagai, Y., et al., 2013. p130Cas, Crk-associated substrate, plays important roles in osteoclastic bone resorption. *J. Bone Miner. Res.* 28 (12), 2449–2462.
- Nakagawa, T., et al., 2008. A tissue biomarker panel predicting systemic progression after PSA recurrence post-definitive prostate cancer therapy. *PLoS One* 3 (5), e2318.
- Nakamoto, T., et al., 2000. Clz, a zinc finger protein that interacts with p130(cas) and activates the expression of matrix metalloproteinases. *Mol. Cell. Biol.* 20 (5), 1649–1658.
- Neumann, L.C., et al., 2011. EFS shows biallelic methylation in uveal melanoma with poor prognosis as well as tissue-specific methylation. *BMC Cancer* 11, 380.
- Nikonova, A.S., et al., 2014a. Nedd9 restrains renal cystogenesis in Pkd1^{−/−} mice. *Proc. Natl. Acad. Sci. U. S. A.* 111 (35), 12859–12864.
- Nikonova, A.S., et al., 2014b. CAS proteins in health and disease: an update. *IUBMB Life* 66 (6), 387–395.
- Ogden, K., et al., 2006. A new signaling paradigm for serotonin: use of Crk-associated substrate in arterial contraction. *Am. J. Physiol. Heart Circ. Physiol.* 291 (6), H2857–H2863.
- Ohba, T., et al., 1998. Dot far-western blot analysis of relative binding affinities of the Src homology 3 domains of Efs and its related proteins. *Anal. Biochem.* 262 (2), 185–192.
- O'Neill, G.M., Fashena, S.J., Golemis, E.A., 2000. Integrin signalling: a new Cas(t) of characters enters the stage. *Trends Cell Biol.* 10 (3), 111–119.
- Pellicena, P., Miller, W.T., 2001. Processive phosphorylation of p130Cas by Src depends on SH3–polyproline interactions. *J. Biol. Chem.* 276 (30), 28190–28196.
- Petch, L.A., et al., 1995. Adhesion-induced tyrosine phosphorylation of the p130 src substrate. *J. Cell Sci.* 108 (Pt 4), 1371–1379.
- Poaty, H., et al., 2012. Genome-wide high-resolution aCGH analysis of gestational chorionicarcinomas. *PLoS One* 7 (1), e29426.
- Poulikakos, P.I., et al., 2011. RAF inhibitor resistance is mediated by dimerization of aberrantly spliced BRAF(V600E). *Nature* 480 (7377), 387–390.
- Pugacheva, E.N., Golemis, E.A., 2005. The focal adhesion scaffolding protein HEF1 regulates activation of the Aurora-A and Nek2 kinases at the centrosome. *Nat. Cell Biol.* 7 (10), 937–946.
- Pugacheva, E.N., et al., 2007. HEF1-dependent Aurora A activation induces disassembly of the primary cilium. *Cell* 129 (7), 1351–1363.
- Roe, K., et al., 2013. Hypoxic tumor kinase signaling mediated by STAT5A in development of castration-resistant prostate cancer. *PLoS One* 8 (5), e63723.
- Rosenthal, S.L., et al., 2014. Connecting the dots: potential of data integration to identify regulatory SNPs in late-onset Alzheimer's disease GWAS findings. *PLoS One* 9 (4), e95152.
- Ruiz, A., et al., 2014. Follow-up of loci from the International Genomics of Alzheimer's Disease Project identifies TRIP4 as a novel susceptibility gene. *Transl. Psychiatry* 4, e358.
- Sakai, R., et al., 1994a. Characterization, partial purification, and peptide sequencing of p130, the main phosphoprotein associated with v-Crk oncoprotein. *J. Biol. Chem.* 269 (52), 32740–32746.
- Sakai, R., et al., 1994b. A novel signaling molecule, p130, forms stable complexes in vivo with v-Crk and v-Src in a tyrosine phosphorylation-dependent manner. *EMBO J.* 13 (16), 3748–3756.
- Sawada, Y., et al., 2006. Force sensing by mechanical extension of the Src family kinase substrate p130Cas. *Cell* 127 (5), 1015–1026.
- Sertkaya, S., et al., 2015. Decreased expression of EFS is correlated with the advanced prostate cancer. *Tumour Biol.* 36 (2), 799–805.
- Shi, J., Casanova, J.E., 2006. Invasion of host cells by *Salmonella typhimurium* requires focal adhesion kinase and p130Cas. *Mol. Biol. Cell* 17 (11), 4698–4708.
- Singh, M.K., et al., 2008. A novel Cas family member, HEPL, regulates FAK and cell spreading. *Mol. Biol. Cell* 19 (4), 1627–1636.
- Songyang, Z., et al., 1993. SH2 domains recognize specific phosphopeptide sequences. *Cell* 72 (5), 767–778.
- Stappenbeck, T.S., et al., 2011. Crohn disease: a current perspective on genetics, autophagy and immunity. *Autophagy* 7 (4), 355–374.
- Stofega, M.R., et al., 1997. Activation of extracellular signal-regulated kinase (ERK) by mitogenic stimuli is repressed in v-Src-transformed cells. *Cell Growth Differ.* 8 (1), 113–119.
- Tachibana, K., et al., 1997. Tyrosine phosphorylation of Crk-associated substrates by focal adhesion kinase. A putative mechanism for the integrin-mediated tyrosine phosphorylation of Crk-associated substrates. *J. Biol. Chem.* 272 (46), 29083–29090.
- Tahiri, A., et al., 2013. Differential inhibition of ex-vivo tumor kinase activity by vemurafenib in BRAF(V600E) and BRAF wild-type metastatic malignant melanoma. *PLoS One* 8 (8), e72692.
- Tazaki, Y., et al., 2010. p130Cas, Crk-associated substrate plays essential roles in liver development by regulating sinusoidal endothelial cell fenestration. *Hepatology* 52 (3), 1089–1099.
- Tchemev, V.T., et al., 2002. The Chediak–Higashi protein interacts with SNARE complex and signal transduction proteins. *Mol. Med.* 8 (1), 56–64.
- Tikhmyanova, N., Little, J.L., Golemis, E.A., 2010a. CAS proteins in normal and pathological cell growth control. *Cell. Mol. Life Sci.* 67 (7), 1025–1048.
- Tikhmyanova, N., et al., 2010b. Dcas supports cell polarization and cell–cell adhesion complexes in development. *PLoS One* 5 (8), e12369.
- Tong, Y., et al., 2012. Genomic characterization of human and rat prolactinomas. *Endocrinology* 153 (8), 3679–3691.
- Tornillo, G., Defilippi, P., Cabodi, S., 2014. Cas proteins: dodgy scaffolding in breast cancer. *Breast Cancer Res.* 16 (5), 443.
- UCSC, C. Bioinformatics Genome Browser <http://www.sabiosciences.com/chipqcrsearch.php>.
- van Agthoven, T., et al., 2009. Relevance of breast cancer antiestrogen resistance genes in human breast cancer progression and tamoxifen resistance. *J. Clin. Oncol.* 27 (4), 542–549.
- Vanaja, D.K., et al., 2009. Hypermethylation of genes for diagnosis and risk stratification of prostate cancer. *Cancer Investig.* 27 (5), 549–560.
- Vital, A.L., et al., 2010. Gene expression profiles of human glioblastomas are associated with both tumor cytogenetics and histopathology. *Neuro Oncol.* 12 (9), 991–1003.
- Wallez, Y., et al., 2012. NSP–CAS protein complexes: emerging signaling modules in cancer. *Genes Cancer* 3 (5–6), 382–393.
- Wang, X., et al., 2015. Genetic determinants of disease progression in Alzheimer's disease. *J. Alzheimers Dis.* 43 (2), 649–655.
- Weidow, C.L., et al., 2000. CAS/Crk signalling mediates uptake of Yersinia into human epithelial cells. *Cell. Microbiol.* 2 (6), 549–560.
- Wright, F.A., et al., 2011. Genome-wide association and linkage identify modifier loci of lung disease severity in cystic fibrosis at 11p13 and 20q13.2. *Nat. Genet.* 43 (6), 539–546.
- Wu, C., et al., 2007. Systematic identification of SH3 domain-mediated human protein–protein interactions by peptide array target screening. *Proteomics* 7 (11), 1775–1785.
- Xing, L., et al., 2000. c-Src signaling induced by the adapters Sin and Cas is mediated by Rap1 GTPase. *Mol. Cell. Biol.* 20 (19), 7363–7377.
- Xing, L., et al., 2004. The adapter molecule Sin regulates T-cell-receptor-mediated signal transduction by modulating signaling substrate availability. *Mol. Cell. Biol.* 24 (10), 4581–4592.
- Yang, L.T., Alexandropoulos, K., Sap, J., 2002. c-SRC mediates neurite outgrowth through recruitment of Crk to the scaffolding protein Sin/Efs without altering the kinetics of ERK activation. *J. Biol. Chem.* 277 (20), 17406–17414.
- Zahedi, R.P., et al., 2008. Phosphoproteome of resting human platelets. *J. Proteome Res.* 7 (2), 526–534.
- Zheng, X.M., Resnick, R.J., Shalloway, D., 2000. A phosphotyrosine displacement mechanism for activation of Src by PTPalpha. *EMBO J.* 19 (5), 964–978.
- Zimman, A., et al., 2014. Phosphoproteomic analysis of platelets activated by pro-thrombotic oxidized phospholipids and thrombin. *PLoS One* 9 (1), e84488.